

Database Schema

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Accessibility Notice

This document is also available in HTML format at:

Link: <https://aholdengouveia.name/AdvData/labs/week4schema.html>

The HTML version provides enhanced accessibility features including keyboard navigation, screen reader support, responsive design, dark mode support, and high contrast options.

Objectives:

- Learn how to evaluate which schema is appropriate
- Create your own schema

Complete the following problems

References, a video, a PowerPoint and some notes are available at my website

Link: <https://www.aholdengouveia.name/AdvData/databaseschema.html>

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Pick a topic you're interested in. This could be anything; sports, collectable card games, cars, video games, etc.

Decide on a Schema

- Out of the 6 design option, pick 3 to evaluate, list them here
- For each design option, give 2 pros and 2 cons for your data specifically
- List out some potential data options for your dataset For example, books might have authors, publication dates and publishers, genre, sub genres, purchase types such as audio or ebook, part of a series or world, connections to other books, etc.
- Make a list of what you think some potential tables could be, include a primary key for each table
- Make notes on how the tables might relate to each other using example, list out potential foreign keys, if you have no foreign keys, explain why not.
- Pick which one you think would be best for your given topic and explain why you chose it in one paragraph or less. Include your pros and cons.

Create your schema

The tool that I used is Link: <https://app.diagrams.net/> You may use other tools of your choice, but make sure everything is clearly labelled and the image includes your name and term somewhere.

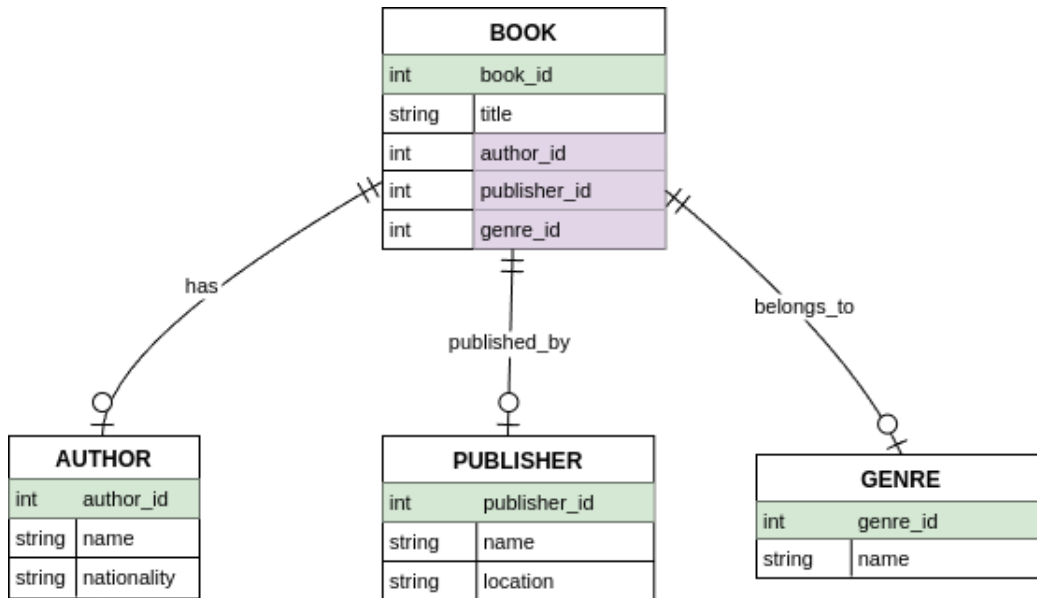
See the linked schema examples Link: <https://www.aholdengouveia.name/AdvData/labs/cis210schemaexampleV1.png> and Link: <https://www.aholdengouveia.name/AdvData/labs/cis210schemaexampleV2.png> for an idea of what you should be creating, don't copy it or feel you have to use the same type, pick the one that works best for your data. My example does NOT include all the information you should have, you should have more tables, and a key, maybe some relationship tables instead of short notes, and rules listed out. You are also likely to have more relationship types with your data, I kept my example small so you'd have an idea of what I'm looking for.

- Make sure to include all your tables, there should be at least 4.
- Each table needs to have a labeled primary key.
- You may need to include foreign keys, if you do then label them.

Deliverables

- Document that includes answers to the above questions for deciding on which schema to use

- Document/image of your schema following the above guidelines.



Green is the Primary Key

Purple is the foreign key

Figure 1: This is an example of a schema. My example does NOT include all the information you should have, you should have more tables, and a key, maybe some relationship tables instead of short notes, and rules listed out.